

Image and Video Generations

Lecture 8: Inverse Problems

劉育綸

Yu-Lun (Alex) Liu



Recall: Bayes' Rule

Bayes' rule is a mathematical formula used to determine the **conditional probability** of events.

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{z}|\mathbf{x})p(\mathbf{z})}{p(\mathbf{x})}$$

$$p(\mathbf{z}|\mathbf{x})p(\mathbf{z}) = p(\mathbf{z}|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}, \mathbf{z})$$

Recall: Marginal Distribution

The **marginal** distribution of a **subset** of a set of random variables is the probability distribution of the variables contained in the subset.

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z}$$


Take the integral over $\mathbf{z}$ to marginalize $\mathbf{x}$.

Recall: Gaussians and Score Function

Let $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{x} = \boldsymbol{\mu} + \Sigma^{\frac{1}{2}}\mathbf{z}$. Then,

- $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.
- $\log p(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) + c$
- $\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$

Recall: Forward Jump

Let $\mathbf{x}_t$ be the noisy data point at time step t from $\mathbf{x}_0$ in the diffusion process.

$$p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

Recall: Tweedie's Formula

How to estimate the **true mean** of a normal distribution from samples drawn from it?

For a $\mathbf{x} \sim p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\mathbb{E}[\boldsymbol{\mu}|\mathbf{x}] = \mathbf{x} + \boldsymbol{\Sigma} \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

給定觀測值 $\mathbf{x}$ ，真實平均值 $\boldsymbol{\mu}$ 的 posterior 期望值等於**觀測值加上一個修正項**

第一項 $\mathbf{x}$ ：我們的起點是 noisy latent 觀測值

第二項 $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ ：這是一個 **denoise 修正項**

- $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ 是 score function
- 他指向**機率密度增加的方向**
- $\boldsymbol{\Sigma}$ 調整修正的幅度

$\mathbf{x}$ 在高密度區域： $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ 較小 $\rightarrow$ 修正較少
 $\mathbf{x}$ 在低密度區域： $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ 較大 $\rightarrow$ 修正較多
這自然地將觀測值「拉向」分布的高密度區域

Recall: Tweedie's Formula

How to estimate the **true mean** of a normal distribution from samples drawn from it?

For a $\mathbf{x} \sim p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\mathbb{E}[\boldsymbol{\mu}|\mathbf{x}] = \mathbf{x} + \boldsymbol{\Sigma} \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

DDPM 的反向去噪過程本質上就是重複應用 Tweedie's Formula:

1. 在每個 timestep t ，我們有 noisy latent $\mathbf{x}_t$
2. 使用學到的 score function 估計 denoise 方向
3. Iteratively 去噪恢復原始影像

Quiz 1-3

Q1. Express $p(\mathbf{y})$ with respect to $p(\mathbf{y}|\mathbf{x})$ and $p(\mathbf{x})$.

Answer 1

A1.

$$p(\mathbf{y}) = \int p(\mathbf{y}, \mathbf{x}) d\mathbf{x} \quad // \text{ Marginalization}$$

$$= \int p(\mathbf{y}|\mathbf{x})p(\mathbf{x})d\mathbf{x} \quad // \text{ Bayes' Rule}$$

Quiz 1-3

Q1. Express $p(\mathbf{y})$ with respect to $p(\mathbf{y}|\mathbf{x})$ and $p(\mathbf{x})$.

Q2. Express $p(\mathbf{y}|\mathbf{x}_t)$ with respect to $p(\mathbf{y}|\mathbf{x}_0)$ and $p(\mathbf{x}_0|\mathbf{x}_t)$.

Note that $p(\mathbf{y}|\mathbf{x}_0, \mathbf{x}_t) = p(\mathbf{y}|\mathbf{x}_0)$.

Since $\mathbf{x}_t$ is sample from $\mathbf{x}_0$, $\mathbf{x}_t$ does not provide any additional information about $\mathbf{y}$.

Answer 2

A2.

$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0)d\mathbf{x}_0$$

Add $\mathbf{x}_t$ conditioning:

Since $\mathbf{x}_t$ is sample from $\mathbf{x}_0$,
 $\mathbf{x}_t$ does not provide any additional information about $\mathbf{y}$.

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}_t) &= \int p(\mathbf{y}|\mathbf{x}_0, \cancel{\mathbf{x}_t})p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0 \\ &= \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0 \end{aligned}$$

Quiz 1-3

Q1. Express $p(\mathbf{y})$ with respect to $p(\mathbf{y}|\mathbf{x})$ and $p(\mathbf{x})$.

Q2. Express $p(\mathbf{y}|\mathbf{x}_t)$ with respect to $p(\mathbf{y}|\mathbf{x}_0)$ and $p(\mathbf{x}_0|\mathbf{x}_t)$.

Note that $p(\mathbf{y}|\mathbf{x}_0, \mathbf{x}_t) = p(\mathbf{y}|\mathbf{x}_0)$.

Since $\mathbf{x}_t$ is sample from $\mathbf{x}_0$, $\mathbf{x}_t$ does not provide any additional information about $\mathbf{y}$.

Q3. Given a sample $\mathbf{x}_t$, what is the expected value of $\mathbf{x}_0$, $\mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t]$?

Answer 3

A3.

$$p(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

From Tweedie's formula:

For a $\mathbf{x} \sim p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\mathbb{E}[\boldsymbol{\mu} | \mathbf{x}] = \mathbf{x} + \boldsymbol{\Sigma} \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

$$\mathbb{E}[\sqrt{\bar{\alpha}_t} \mathbf{x}_0 | \mathbf{x}_t] = \sqrt{\bar{\alpha}_t} \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0)$$

$$\therefore \mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0))$$

給定 noisy latent $\mathbf{x}_t$ ，我們可以用 Tweedie's formula 估計原始影像 $\mathbf{x}_0$

Answer 3

A3.

$$p(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

From Tweedie's formula:

For a $\mathbf{x} \sim p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\mathbb{E}[\boldsymbol{\mu} | \mathbf{x}] = \mathbf{x} + \boldsymbol{\Sigma} \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

$$\mathbb{E}[\sqrt{\bar{\alpha}_t} \mathbf{x}_0 | \mathbf{x}_t] = \sqrt{\bar{\alpha}_t} \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0)$$

$$\therefore \mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0))$$

第一項 $\frac{\mathbf{x}_t}{\sqrt{\bar{\alpha}_t}}$: 是簡單的對 $\mathbf{x}_t$ 縮放

給定 noisy latent $\mathbf{x}_t$, 我們可以用 Tweedie's formula 估計原始影像 $\mathbf{x}_0$

Answer 3

A3.

$$p(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

From Tweedie's formula:

For a $\mathbf{x} \sim p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\mathbb{E}[\boldsymbol{\mu} | \mathbf{x}] = \mathbf{x} + \boldsymbol{\Sigma} \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

$$\mathbb{E}[\sqrt{\bar{\alpha}_t} \mathbf{x}_0 | \mathbf{x}_t] = \sqrt{\bar{\alpha}_t} \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0)$$

第二項 $\frac{(1 - \bar{\alpha}_t)}{\sqrt{\bar{\alpha}_t}} \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0)$: 是 denoise 修正項

$$\therefore \mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log p(\mathbf{x}_t | \mathbf{x}_0))$$

給定 noisy latent $\mathbf{x}_t$, 我們可以用 Tweedie's formula 估計原始影像 $\mathbf{x}_0$

Summary

$$p(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0$$

$$\mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t] = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla_x \log p(\mathbf{x}_t|\mathbf{x}_0))$$

給定 noisy latent $\mathbf{x}_t$ ，我們可以用 Tweedie's formula 估計原始影像 $\mathbf{x}_0$

Quiz 4

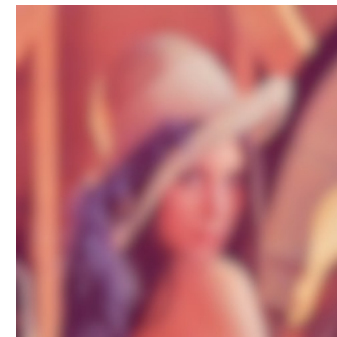
- Let $\mathbf{y} = A\mathbf{x}_0 + \sigma\mathbf{z}$ where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then,

$$p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2\mathbf{I})$$

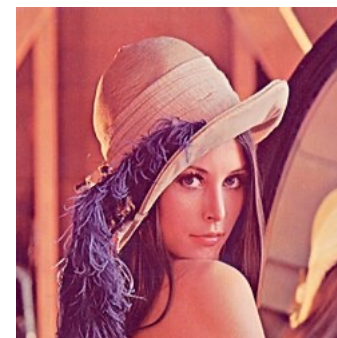
- Let $\mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t]$ and $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2\mathbf{I})$.

- $p(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0) p(\mathbf{x}_0|\mathbf{x}_t) d\mathbf{x}_0$

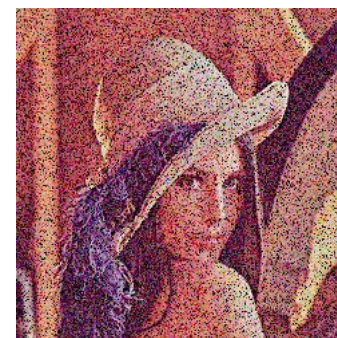
Q4. What is $p(\mathbf{y}|\mathbf{x}_t)$?



觀測到被破壞的影像 $\mathbf{y}$



原始沒被破壞前的正確答案 $\mathbf{x}_0$



在 timestep t 的 noisy latent $\mathbf{x}_t$

Recall: Marginal and Conditional Gaussians

When $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \sigma_x^2 \mathbf{I})$ and $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(A\mathbf{x} + b, \sigma_y^2 \mathbf{I})$,

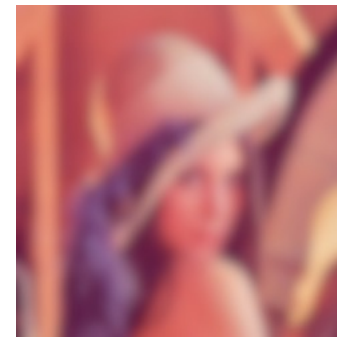
$$\underbrace{p(\mathbf{y})}_{\text{Marginal}} = \int \underbrace{p(\mathbf{y}|\mathbf{x})}_{\text{Likelihood}} \underbrace{p(\mathbf{x})}_{\text{Prior}} d\mathbf{x} = \mathcal{N}(A\boldsymbol{\mu} + b, (\sigma_y^2 + \sigma_x^2 AA^T)\mathbf{I})$$

Answer 4

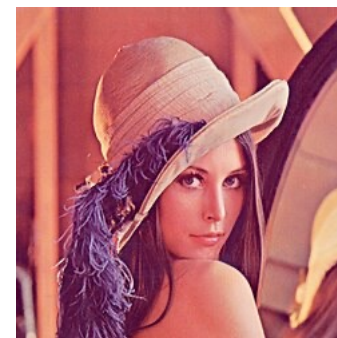
$$\mathbf{A4.} \quad p(\mathbf{y}|\mathbf{x}_t) = \int \underbrace{p(\mathbf{y}|\mathbf{x}_0)}_{\text{Likelihood}} \underbrace{p(\mathbf{x}_0|\mathbf{x}_t)}_{\text{Prior}} d\mathbf{x}_0$$

- Prior $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2\mathbf{I})$
- Likelihood $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2\mathbf{I})$

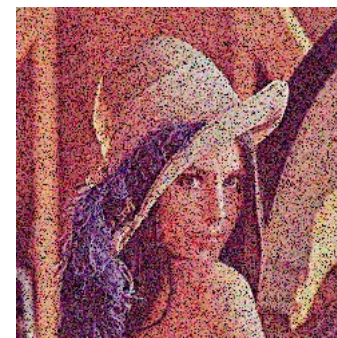
$$\therefore p(\mathbf{y}|\mathbf{x}_t) = \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2\mathbf{I} + r^2AA^T)$$



觀測到被破壞的影像 $\mathbf{y}$



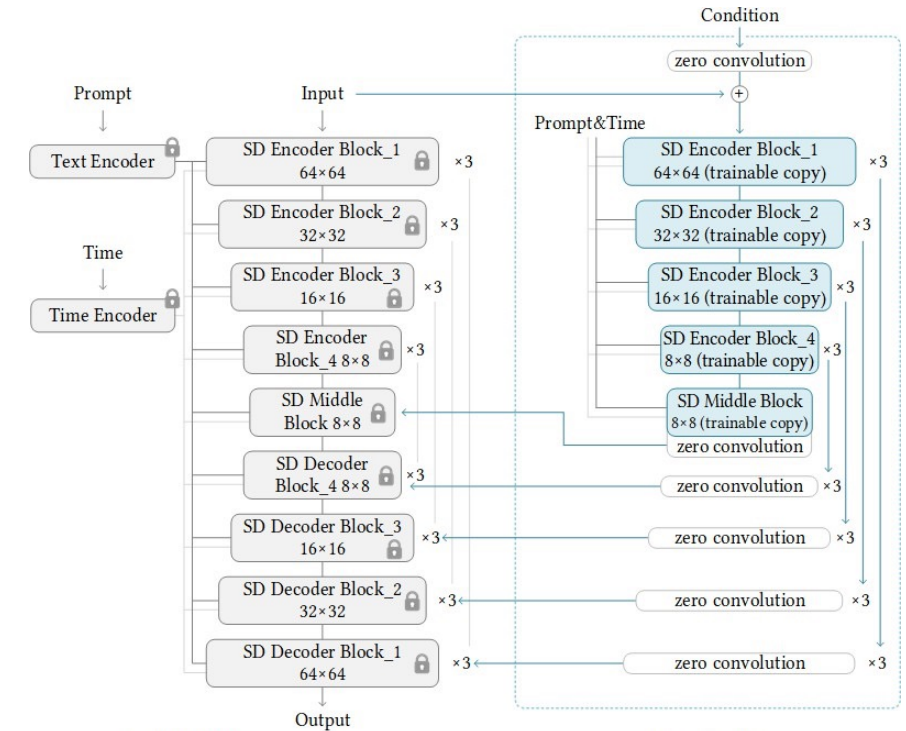
原始沒被破壞前的正確答案 $\mathbf{x}_0$



在 timestep t 的 noisy latent $\mathbf{x}_t$

Finetuning for Conditional Generation

ControlNet

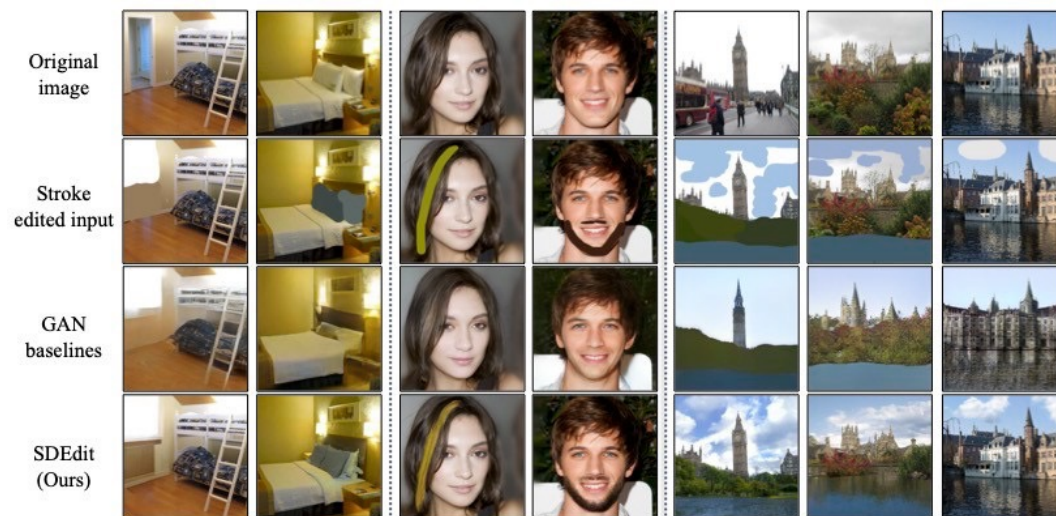
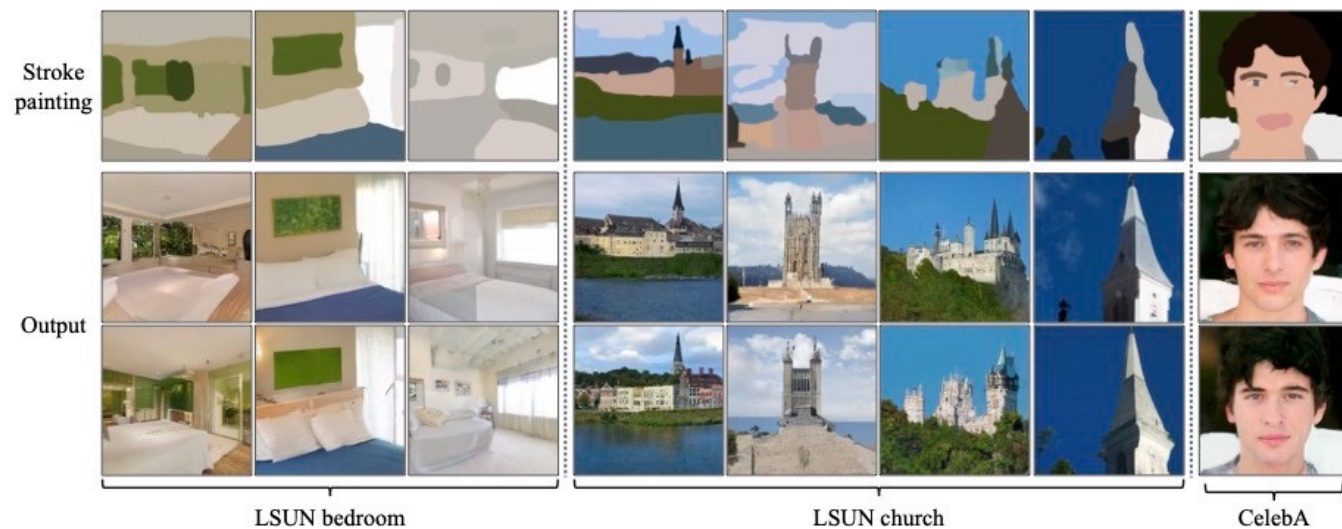


(a) Stable Diffusion

(b) ControlNet

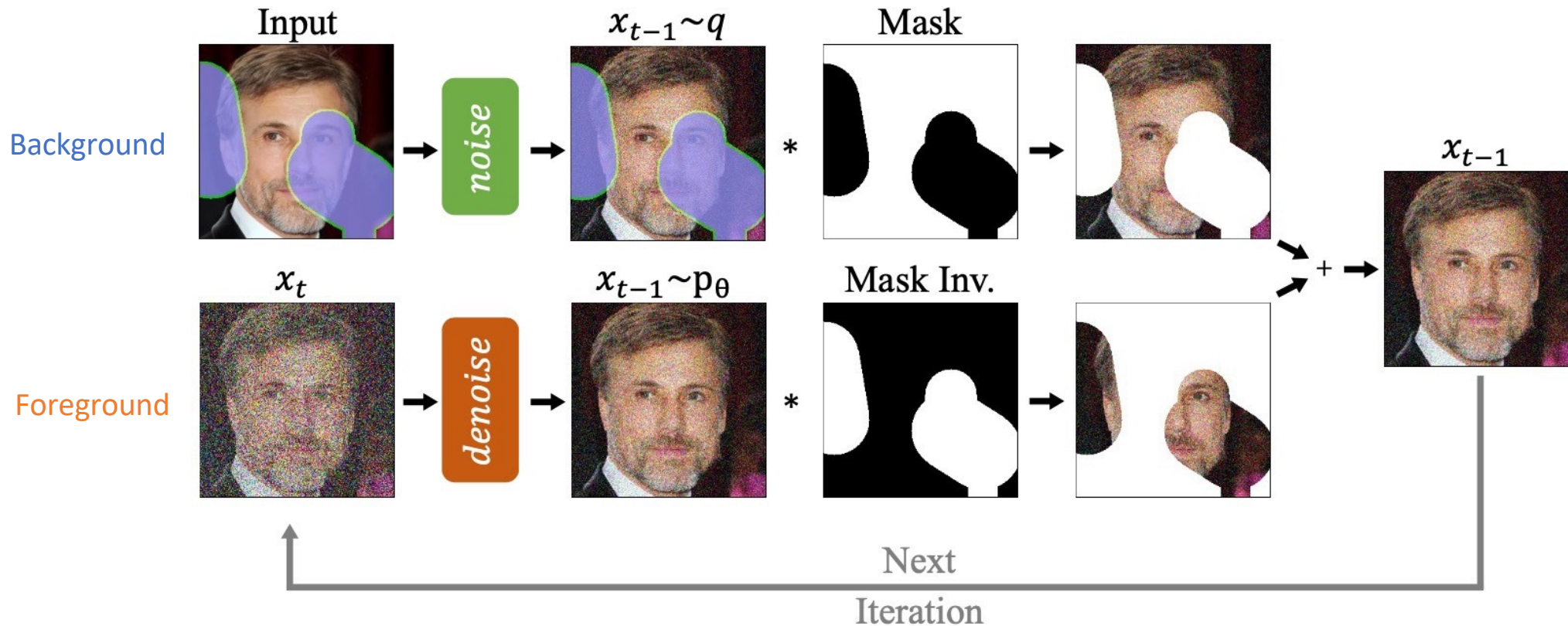
Training-Free Conditional Generation

SDEdit



Training-Free Conditional Generation

RePaint



Inverse Problems

Find a data point (causal factor) $\mathbf{x}$ that corresponds to the measurement (observation) $\mathbf{y}$ given the following corruption model:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \sigma \mathbf{z} \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Q. 如果 $\mathcal{A}$ 只是一個 linear function，例如矩陣乘法
那我們可以直接 inverse 矩陣 $\mathcal{A}$ 來解這個問題？

Inverse Problems

Find a data point (causal factor) $\mathbf{x}$ that corresponds to the measurement (observation) $\mathbf{y}$ given the following corruption model:

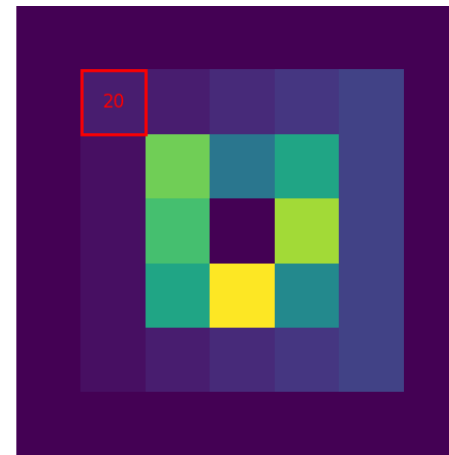
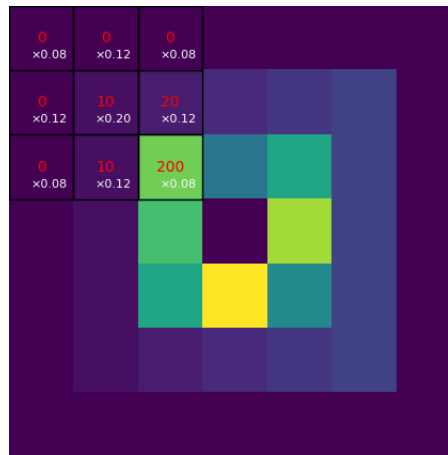
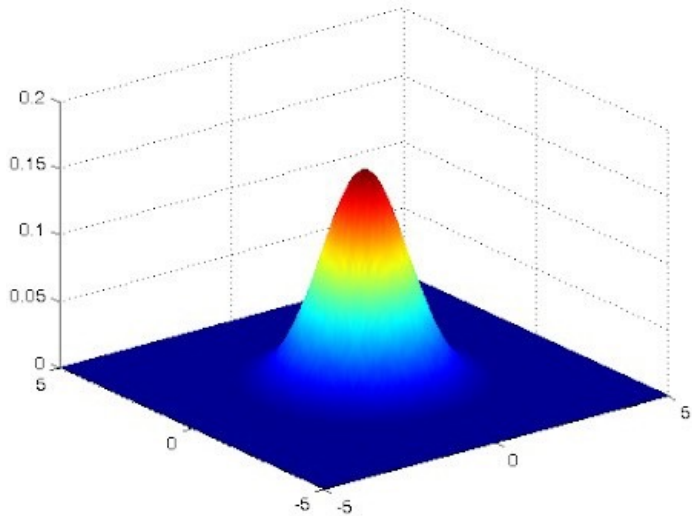
$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \sigma \mathbf{z} \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

A.

- $\mathcal{A}$ 通常不可逆，例如影像的 downsampling
- 就算 $\mathcal{A}$ 可逆，直接用 inverse 求解會放大 $\mathbf{z}$ 這個 noise
- Ill-posed problem：可能的 $\mathbf{x}$ 有無窮多組解（思考 super-resolution 的情況）

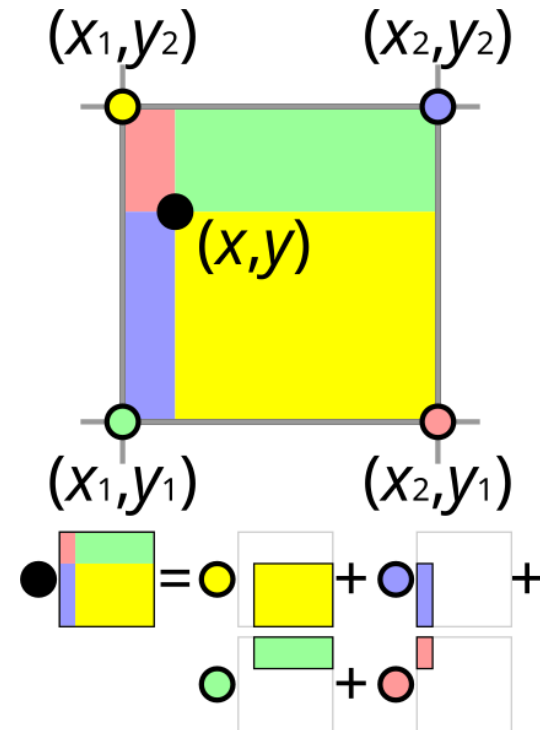
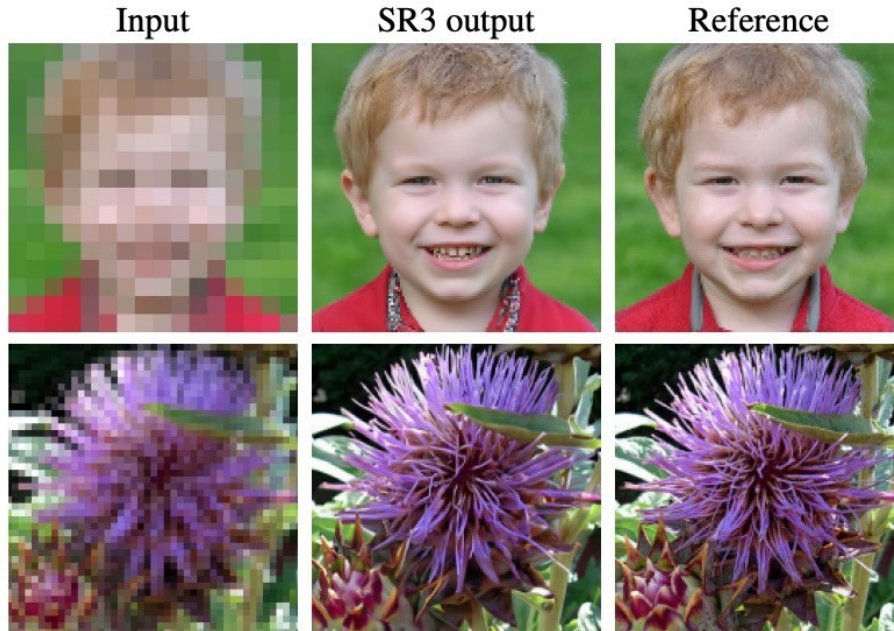
Example: Gaussian Deblurring

- y : a blurry image.
- $\mathcal{A}$: the convolution with a Gaussian kernel.



Example: Super-Resolution

- y : a low-resolution image.
- $\mathcal{A}$: the downsampling function (e.g., bilinear or bicubic interpolation).



Example: Inpainting/Outpainting

- y : a low-resolution image.

Q. What should be $\mathcal{A}$ in inpainting/outpainting cases?



Example: Inpainting/Outpainting

- y : a low-resolution image.
- $\mathcal{A}$: the masking function.



Inverse Problems

- The solution to an inverse problem may not be **unique**.
- **Conditional generative models** are needed.



Inverse Problems

跟 predict noise 是等價的

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) = -\frac{\hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}}$$

- Given a pretrained diffusion model learned the score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$, how to compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$?

也就是在每一個 denoising 的 iteration 都考慮 $\mathbf{y}$ 的 condition

- From Bayes' rule:

$$p(\mathbf{x}_t | \mathbf{y}) = \frac{p(\mathbf{x}_t)p(\mathbf{y}|\mathbf{x}_t)}{\int_{\mathbf{x}_t} p(\mathbf{x}_t)p(\mathbf{y}|\mathbf{x}_t)d\mathbf{x}_t}$$

Constant

$$\therefore \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

Inverse Problems

When $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ is given, we can compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$ from $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$.

Q. We previously discussed how to compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$ before. How?

Inverse Problems

When $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ is given, we can compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$ from $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$.

A. Classifier guidance and classifier-free guidance

Recall: Classifier Guidance

How to compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$?

- Train a **classifier** $p_\phi(\mathbf{y}|\mathbf{x}_t)$, taking noisy data $\mathbf{x}_t$ (and time step t) as input and classifying it.
- Take the gradient of the classifier $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$

Recall: Classifier-Free Guidance

How to compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$?

Train a noise prediction network $\hat{\epsilon}_\theta$ that learns both the unconditional and conditional noise:

- Unconditional noise: $\hat{\epsilon}_\theta(\mathbf{x}_t, \phi, t) = -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$
- Conditional noise: $\hat{\epsilon}_\theta(\mathbf{x}_t, \mathbf{y}, t) = -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$

Recall: Classifier-Free Guidance

Since

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

Combine both noise as follows ($\lambda = 1$ case):

$$\tilde{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) = \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) + \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \phi, t)$$

缺點：需要有 conditional input / output 的 training data

Recall: Classifier-Free Guidance

How to compute $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$?

Train a noise prediction network $\hat{\epsilon}_\theta$ that learns both the unconditional and conditional noise:

- Unconditional noise: $\hat{\epsilon}_\theta(\mathbf{x}_t, \phi, t) = -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$
- Conditional noise: $\hat{\epsilon}_\theta(\mathbf{x}_t, \mathbf{y}, t) = -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$

Q. 能不能在完全不 fine-tune 的情況下，解 inverse problem 呢？

How to model $\nabla_{x_t} \log p(\mathbf{y}|\mathbf{x}_t)$
without training?

By just adding one line of code in the denoising process.

Pseudoinverse-Guided Diffusion Models (PGDM)

Consider a special case first, assuming a **linear** function:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma \mathbf{z} = A\mathbf{x}_0 + \sigma \mathbf{z}$$

In most cases, $\mathcal{A}$ is a **linear** function:

- Convolution
- Interpolation
- Masking
- Projection

Pseudoinverse-Guided Diffusion Models (PGDM)

Consider a special case first, assuming a **linear** function:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma \mathbf{z} = A\mathbf{x}_0 + \sigma \mathbf{z}$$

Recall **Quiz 4**:

- $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2 \mathbf{I})$
- Let $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$ with variance **r^2** . Then,

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2 \mathbf{I} + r^2 AA^T)$$

Recall: Classifier-Free Guidance

Since

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

Combine both noise as follows ($\lambda = 1$ case):

$$\tilde{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) = \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) + \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, \phi, t)$$

Pseudoinverse-Guided Diffusion Models (PGDM)

Consider a special case first, assuming a **linear** function:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma \mathbf{z} = A\mathbf{x}_0 + \sigma \mathbf{z}$$

Recall **Quiz 4**:

- $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2 \mathbf{I})$
- Let $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$ with variance **r^2** . Then,

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2 \mathbf{I} + r^2 AA^T)$$

Pseudoinverse-Guided Diffusion Models (Π GDM)

- Let $\Sigma = \sigma^2 \mathbf{I} + r^2 AA^T$.

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \Sigma).$$

- Also recall that when $p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \Sigma)$,

$$\log p(\mathbf{y}|\mathbf{x}_t) = -\frac{1}{2}(\mathbf{y} - A\mathbf{x}_{0|t})^T \Sigma^{-1}(\mathbf{y} - A\mathbf{x}_{0|t}) + c.$$

Pseudoinverse-Guided Diffusion Models (PGDM)

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &= \frac{\partial \log p(\mathbf{y}|\mathbf{x}_t)}{\partial \mathbf{x}_{0|t}} \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t} \\ &\approx \frac{\partial}{\partial \mathbf{x}_{0|t}} \left(-\frac{1}{2} (\mathbf{y} - A\mathbf{x}_{0|t})^T \Sigma^{-1} (\mathbf{y} - A\mathbf{x}_{0|t}) \right) \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t} \\ &= A^T \Sigma^{-1} (\mathbf{y} - A\mathbf{x}_{0|t}) \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t} \\ &= A^T (\sigma^2 \mathbf{I} + r^2 A A^T)^{-1} \boxed{(\mathbf{y} - A\mathbf{x}_{0|t})} \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t}\end{aligned}$$

觀測 $\mathbf{y}$ 與經過 A 破壞的 $\mathbf{x}_{0|t}$ prediction 之間的差異

Pseudoinverse-Guided Diffusion Models (PGDM)

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \approx A^T (\sigma^2 \mathbf{I} + r^2 A A^T)^{-1} \boxed{\mathbf{y} - A \mathbf{x}_{0|t}} \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t}$$

觀測 $\mathbf{y}$ 與經過 A 破壞的 $\mathbf{x}_{0|t}$ prediction 之間的差異

Pseudoinverse-Guided Diffusion Models (PGDM)

Consider a special case first, assuming a **linear** function:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma \mathbf{z} = A\mathbf{x}_0 + \sigma \mathbf{z}$$

Recall **Quiz 4**:

- $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2 \mathbf{I})$
- Let $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$ with variance **r^2** . Then,

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2 \mathbf{I} + r^2 AA^T)$$

Pseudoinverse-Guided Diffusion Models (PGDM)

If $\sigma = 0$ (no noise in the measurement),

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \approx \frac{1}{r^2} \underbrace{A^T (AA^T)^{-1}}_{\text{Pseudoinverse of } A} (\mathbf{y} - A\mathbf{x}_{0|t}) \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t}$$

⇒ Pseudoinverse guidance (PG)!

Pseudoinverse-Guided Diffusion Models (PGDM)

Consider a special case first, assuming a **linear** function:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \sigma \mathbf{z} = A\mathbf{x}_0 + \sigma \mathbf{z}$$

Recall **Quiz 4**:

- $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2 \mathbf{I})$
- Let $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$ with variance **r^2** . Then,

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2 \mathbf{I} + r^2 AA^T)$$

Diffusion Posterior Sampling (DPS)

- What if we assume that $r^2 = 0$?

A distribution with zero variance is a **Dirac delta** function.

- Then,

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2\mathbf{I})$$

- Now we do not need to assume a **linear** function:

$$p(\mathbf{y}|\mathbf{x}_t) \approx \mathcal{N}(\mathcal{A}(\mathbf{x}_{0|t}), \sigma^2\mathbf{I})$$

Diffusion Posterior Sampling (DPS)

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &\approx \nabla_{\mathbf{x}_t} \left(-\frac{\|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2}{2\sigma^2} \right) \\ &= -\frac{1}{2\sigma^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2\end{aligned}$$

從 $\mathbf{x}_t$ 用 Tweedie's formula 預測出 $\mathbf{x}_{0|t}$ ($\mathbf{x}_0$ 的期望值)
用 $\mathcal{A}$ 破壞後，希望與觀測 $\mathbf{y}$ 越像越好
這時的 $\mathbf{x}_t$ 就是我們想要的

Recall: DDPM

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0, \tilde{\sigma}_t^2\mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$

2. $\tilde{\boldsymbol{\mu}} = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_{0|t}$

3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \tilde{\sigma}_t^2\mathbf{z}_t$

Recall: DDPM

$$\begin{aligned} \mathbf{x}_{t-1} &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \left(\frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right) \right) + \tilde{\sigma}_t^2 \mathbf{z}_t \\ &= - \frac{\sqrt{1 - \bar{\alpha}_t} \sqrt{\bar{\alpha}_{t-1}} \beta_t}{\sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_t)} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) + \left(\frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_t)} \right) \mathbf{x}_t + \tilde{\sigma}_t^2 \mathbf{z}_t \end{aligned}$$

修改這個 noise，除了 unconditional score，我們也想加上 conditional score

$\mathbf{x}_{t-1}$ is an **affine function** with respect to $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.

Diffusion Posterior Sampling (DPS)

Let $\mathbf{x}'_{t-1}$ be a sample at time step $t - 1$ with noise $\hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t)$.

Then, what is $\mathbf{x}_{t-1}$ sampled with noise:

$$\hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t) + \boldsymbol{\varepsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t)?$$

Note that $\boldsymbol{\varepsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) = -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$.

Diffusion Posterior Sampling (DPS)

我們想要的 conditional part

$$\downarrow$$
$$\mathbf{x}_{t-1} = \mathbf{x}'_{t-1} - \frac{\sqrt{1 - \bar{\alpha}_t} \sqrt{\bar{\alpha}_{t-1}} \beta_t}{\sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_t)} \left(-\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \right)$$

$$= \mathbf{x}'_{t-1} + \frac{\sqrt{1 - \bar{\alpha}_t} \sqrt{\bar{\alpha}_{t-1}} \beta_t}{\sqrt{\bar{\alpha}_t}} \left(-\frac{1}{2\sigma^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2 \right)$$

$$\begin{array}{c} \nearrow \\ \text{Unconditional part} \end{array} = \mathbf{x}'_{t-1} - \underbrace{\frac{\sqrt{1 - \bar{\alpha}_t} \sqrt{\bar{\alpha}_{t-1}} \beta_t}{2\sigma^2 \sqrt{\bar{\alpha}_t}}}_{\text{Let } \xi_t \text{ Scaling factor}} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2$$

與觀測 $\mathbf{y}$ 的修正項

Recall: DDPM

$$\begin{aligned}\mathbf{x}_{t-1} &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \left(\frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right) \right) + \tilde{\sigma}_t^2 \mathbf{z}_t \\ &= -\frac{\sqrt{1 - \bar{\alpha}_t}\sqrt{\bar{\alpha}_{t-1}}\beta_t}{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_t)} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) + \left(\frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_t)} \right) \mathbf{x}_t + \tilde{\sigma}_t^2 \mathbf{z}_t\end{aligned}$$

$\mathbf{x}_{t-1}$ is an **affine function** with respect to $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.

Diffusion Posterior Sampling (DPS)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0, \tilde{\sigma}_t^2\mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$

2. $\tilde{\boldsymbol{\mu}} = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_{0|t}$

3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \tilde{\sigma}_t^2\mathbf{z}_t$

5. $\mathbf{x}_{t-1} = \mathbf{x}'_{t-1} - \xi_t \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2 \longleftarrow \text{One line of code!}$

在 DDPM 的 denoising process 中加一行 code
就可以做到不需要 finetuning 的 conditional generation

Limitation

- Is $p(\mathbf{x}_0|\mathbf{x}_t) \approx \mathcal{N}(\mathbf{x}_{0|t}, r^2\mathbf{I})$ a good approximation?
- Especially, is it a good approximation if $r = 0$?

Inverse Problems – Literature

TFG: Unified Training-Free Guidance for Diffusion Models

Haotian Ye¹* Haowei Lin^{2*} Jiaqi Han^{1*} Minkai Xu¹ Sheng Liu¹
Yitao Liang² Jianzhu Ma³ James Zou¹
Stefano Ermon¹
¹Stanford University ²Peking University ³Tsinghua University

Abstract

Given an unconditional diffusion model and a predictor for a target property of interest (*e.g.*, a classifier), the goal of training-free guidance is to generate samples with desirable target properties without additional training. Existing methods, though effective in various individual applications, often lack theoretical grounding and rigorous testing on extensive benchmarks. As a result, they could even fail on simple tasks, and applying them to a new problem becomes unavoidably difficult. This paper introduces a novel algorithmic framework encompassing existing methods as special cases, unifying the study of training-free guidance into the analysis of an algorithm-agnostic design space. Via theoretical and empirical investigation, we propose an efficient and effective hyper-parameter searching strategy that can be readily applied to any downstream task. We systematically benchmark across 7 diffusion models on 16 tasks with 40 targets, and improve performance by 8.5% on average. Our framework and benchmark offer a solid foundation for conditional generation in a training-free manner.¹

Inverse Problems – Survey

A Survey on Diffusion Models for Inverse Problems

Giannis Daras¹, Hyungjin Chung², Chieh-Hsin Lai³, Yuki Mitsufuji³,
Jong Chul Ye², Peyman Milanfar⁴, Alexandros G. Dimakis¹, Mauricio Delbracio⁴

¹UT Austin ²KAIST ³Sony AI ⁴Google

Abstract

Diffusion models have become increasingly popular for generative modeling due to their ability to generate high-quality samples. This has unlocked exciting new possibilities for solving inverse problems, especially in image restoration and reconstruction, by treating diffusion models as unsupervised priors. This survey provides a comprehensive overview of methods that utilize pre-trained diffusion models to solve inverse problems without requiring further training. We introduce taxonomies to categorize these methods based on both the problems they address and the techniques they employ. We analyze the connections between different approaches, offering insights into their practical implementation and highlighting important considerations. We further discuss specific challenges and potential solutions associated with using latent diffusion models for inverse problems. This work aims to be a valuable resource for those interested in learning about the intersection of diffusion models and inverse problems.

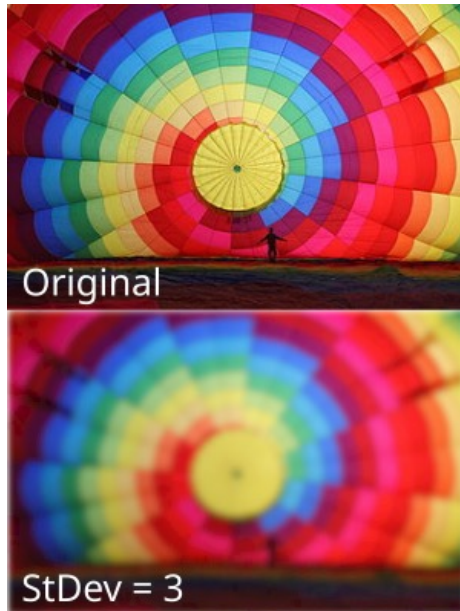
Inverse Problems

Find a data point (causal factor) $\mathbf{x}$ that corresponds to the measurement (observation) $\mathbf{y}$ given the following corruption model:

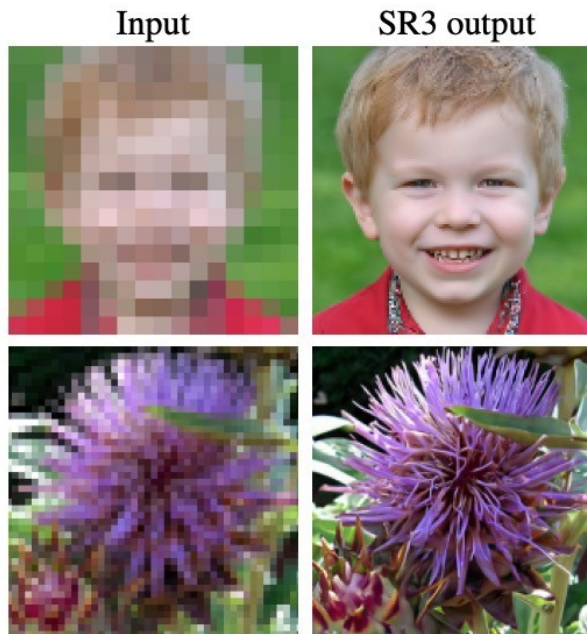
$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \sigma \mathbf{z} \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Inverse Problems

Gaussian Deblurring



Super-Resolution



Inpainting/Outpainting



Inverse Problems

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this. How?

Inverse Problems

$$p(\mathbf{x}_t|\mathbf{y}) = \frac{p(\mathbf{x}_t)p(\mathbf{y}|\mathbf{x}_t)}{\int_{\mathbf{x}_t} p(\mathbf{x}_t)p(\mathbf{y}|\mathbf{x}_t)d\mathbf{x}_t}$$

Constant

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

這項只是 unconditional 的 score

How to model $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$ without training?

Inverse Problems

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this.
- $p(\mathbf{y}|\mathbf{x}_0)$: We know this. What is it?

Inverse Problems

When assuming that $\mathcal{A}(\mathbf{x}_0) = A\mathbf{x}_0$:

$$\mathbf{y} = A\mathbf{x}_0 + \sigma\mathbf{z}$$

$$\therefore p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2\mathbf{I})$$

Inverse Problems

我們真正想要的是 $p(\mathbf{x}_t|\mathbf{y})$
所以我們需要用 $p(\mathbf{y}|\mathbf{x}_t)$ 來算
但我們只有 $p(\mathbf{y}|\mathbf{x}_0)$
所以用 $p(\mathbf{x}_0|\mathbf{x}_t)$ 來橋接

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this.
- $p(\mathbf{y}|\mathbf{x}_0)$: We *only* know this. How to compute $p(\mathbf{y}|\mathbf{x}_t)$?
- $p(\mathbf{x}_0|\mathbf{x}_t)$: Let's approximate this since

$$p(\mathbf{y}|\mathbf{x}_t) = \int \overset{\text{Likelihood}}{p(\mathbf{y}|\mathbf{x}_0)} \overset{\text{Prior}}{p(\mathbf{x}_0|\mathbf{x}_t)} d\mathbf{x}_0$$

Quiz 4

$$p(\mathbf{y}|\mathbf{x}_t) = \int \underbrace{p(\mathbf{y}|\mathbf{x}_0)}_{\text{Likelihood}} \underbrace{p(\mathbf{x}_0|\mathbf{x}_t)}_{\text{Prior}} d\mathbf{x}_0$$

- **Prior** $p(\mathbf{x}_0|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$
- **Likelihood** $p(\mathbf{y}|\mathbf{x}_0) = \mathcal{N}(A\mathbf{x}_0, \sigma^2 \mathbf{I})$

$$\therefore p(\mathbf{y}|\mathbf{x}_t) = \mathcal{N}(A\mathbf{x}_{0|t}, \sigma^2 \mathbf{I} + r^2 AA^T)$$

Inverse Problems

我們真正想要的是 $p(\mathbf{x}_t|\mathbf{y})$
所以我們需要用 $p(\mathbf{y}|\mathbf{x}_t)$ 來算
但我們只有 $p(\mathbf{y}|\mathbf{x}_0)$
所以用 $p(\mathbf{x}_0|\mathbf{x}_t)$ 來橋接

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this.
- $p(\mathbf{y}|\mathbf{x}_0)$: We *only* know this. How to compute $p(\mathbf{y}|\mathbf{x}_t)$?
- $p(\mathbf{x}_0|\mathbf{x}_t)$: How to approximate this?

Inverse Problems

- We know its **mean**: $\mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$.
用 Tweedie's formula 可以從 $\mathbf{x}_t$ 得到 $\mathbf{x}_0$ 的 mean
- Let's assume the **variance** is $r^2 \mathbf{I}$.

$$\therefore p(\mathbf{x}_0 | \mathbf{x}_t) \approx \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$$

Inverse Problems

我們真正想要的是 $p(\mathbf{x}_t|\mathbf{y})$
所以我們需要用 $p(\mathbf{y}|\mathbf{x}_t)$ 來算
但我們只有 $p(\mathbf{y}|\mathbf{x}_0)$
所以用 $p(\mathbf{x}_0|\mathbf{x}_t)$ 來橋接

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this.
- $p(\mathbf{y}|\mathbf{x}_0)$: We *only* know this. How to compute $p(\mathbf{y}|\mathbf{x}_t)$?
- $p(\mathbf{x}_0|\mathbf{x}_t)$: How to approximate this?

只能用 Tweedie's 得到 mean , variance 依然未知

Inverse Problems

- We know its **mean**: $\mathbf{x}_{0|t} = \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$.

用 Tweedie's formula 可以從 $\mathbf{x}_t$ 得到 $\mathbf{x}_0$ 的期望值

- Let's assume the **variance** is $r^2 \mathbf{I}$.

$$\therefore p(\mathbf{x}_0 | \mathbf{x}_t) \approx \mathcal{N}(\mathbf{x}_{0|t}, r^2 \mathbf{I})$$

Inverse Problems

我們真正想要的是 $p(\mathbf{x}_t|\mathbf{y})$
所以我們需要用 $p(\mathbf{y}|\mathbf{x}_t)$ 來算
但我們只有 $p(\mathbf{y}|\mathbf{x}_0)$
所以用 $p(\mathbf{x}_0|\mathbf{x}_t)$ 來橋接

- $p(\mathbf{x}_t|\mathbf{y})$: We need this.
- $p(\mathbf{y}|\mathbf{x}_t)$: We can compute $p(\mathbf{x}_t|\mathbf{y})$ using this.
- $p(\mathbf{y}|\mathbf{x}_0)$: We *only* know this. How to compute $p(\mathbf{y}|\mathbf{x}_t)$?
- $p(\mathbf{x}_0|\mathbf{x}_t)$: How to approximate this?

只能用 Tweedie's 得到 mean
Variance 我們假設成 r^2

Pseudoinverse-Guided Diffusion Models (PGDM)

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

unconditional 的 score

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t) \approx A^T (\sigma^2 \mathbf{I} + r^2 A A^T)^{-1} (\mathbf{y} - A \mathbf{x}_{0|t}) \cdot \frac{\partial \mathbf{x}_{0|t}}{\partial \mathbf{x}_t}$$

長得很像
pseudoinverse 的東西

Tweedie's 推出來的
 $\mathbf{x}_{0|t}$ 經過 A 模擬後跟
真實的觀測 $\mathbf{y}$ 的差距

對 $\mathbf{x}_t$ 算
gradient

Diffusion Posterior Sampling (DPS)

When $r = 0$, 給定 $\mathbf{x}_t$
假設 Tweedie's always sample 出 mean 的期望值
沒有任何 variance 擾動

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &\approx \nabla_{\mathbf{x}_t} \left(-\frac{\|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2}{2\sigma^2} \right) \\ &= -\frac{1}{2\sigma^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2\end{aligned}$$

變得非常簡單：

從當前的 $\mathbf{x}_t$ ，利用 Tweedie's 推出 $\mathbf{x}_{0|t}$ ，經過 $\mathcal{A}$ 的模擬，看看跟真實觀測 $\mathbf{y}$ 差多少，差距以 L2 的形式對 $\mathbf{x}_t$ 計算 gradient 更新 $\mathbf{x}_t$

Diffusion Posterior Sampling (DPS)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0, \tilde{\sigma}_t^2\mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right)$
2. $\tilde{\boldsymbol{\mu}} = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_{0|t}$
3. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
4. $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \tilde{\sigma}_t^2\mathbf{z}_t$
5. $\mathbf{x}_{t-1} = \mathbf{x}'_{t-1} - \xi_t \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbf{x}_{0|t})\|^2$

實作非常簡單：前面都跟原本的 DDPM/DDIM 一樣
只要多加這一個額外的 score 修正項 $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$

這是 Diffusion models 最強大的能力之一！可以用任何 pretrained **unconditional** diffusion，在 reverse process 中加上 score 修正項做到 **guided** generation

Diffusion Posterior Sampling (DPS)

The same idea can be applied for **any guidance function** (loss function) $\mathcal{L}(\boldsymbol{x})$:

$$\boldsymbol{x}_{t-1} = \boldsymbol{x}'_{t-1} - \xi_t \nabla_{\boldsymbol{x}_t} \mathcal{L}(\boldsymbol{x})$$

Loss function 不一定要是 L2 !

只要你能寫出可微分的 **loss** , 都可以用來引導生成

Example: Style Guidance

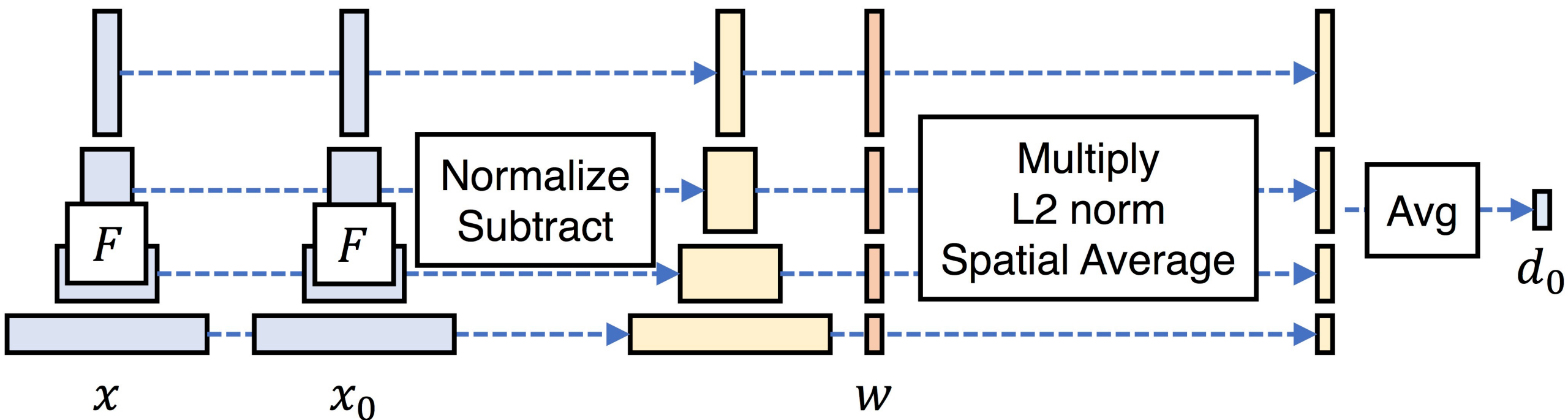
- $\mathcal{L}$: style similarity with the other image.



Off-Topic: Learned Perceptual Image Patch Similarity (LPIPS)

也不一定要 LPIPS，也可以是 VGG, DINO, etc.

Style Similarity Metrics



Example: Coherent Panoramas

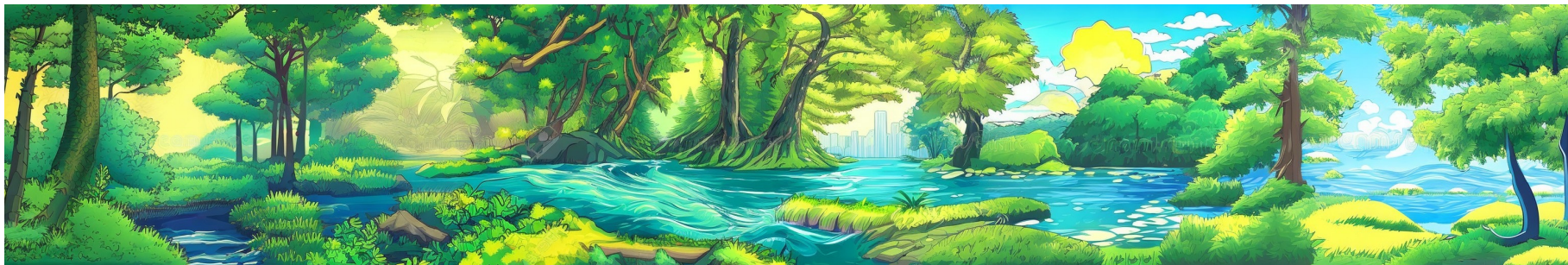
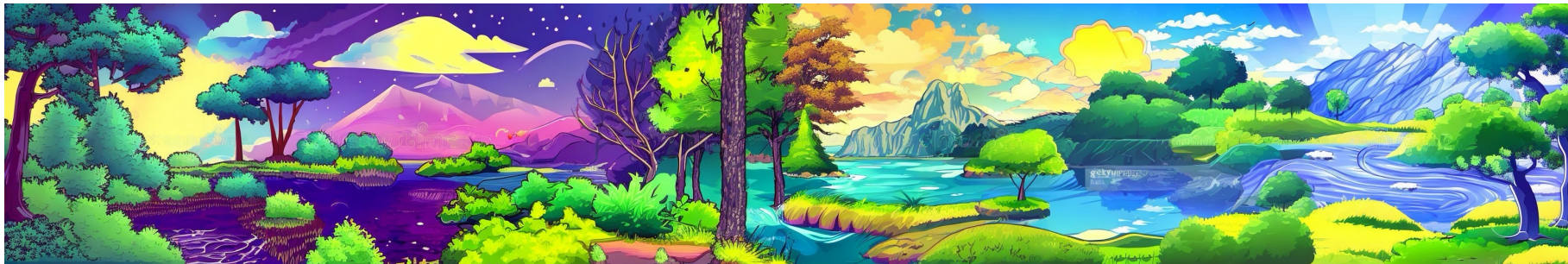
Let's revisit the MultiDiffusion results from the last lecture.



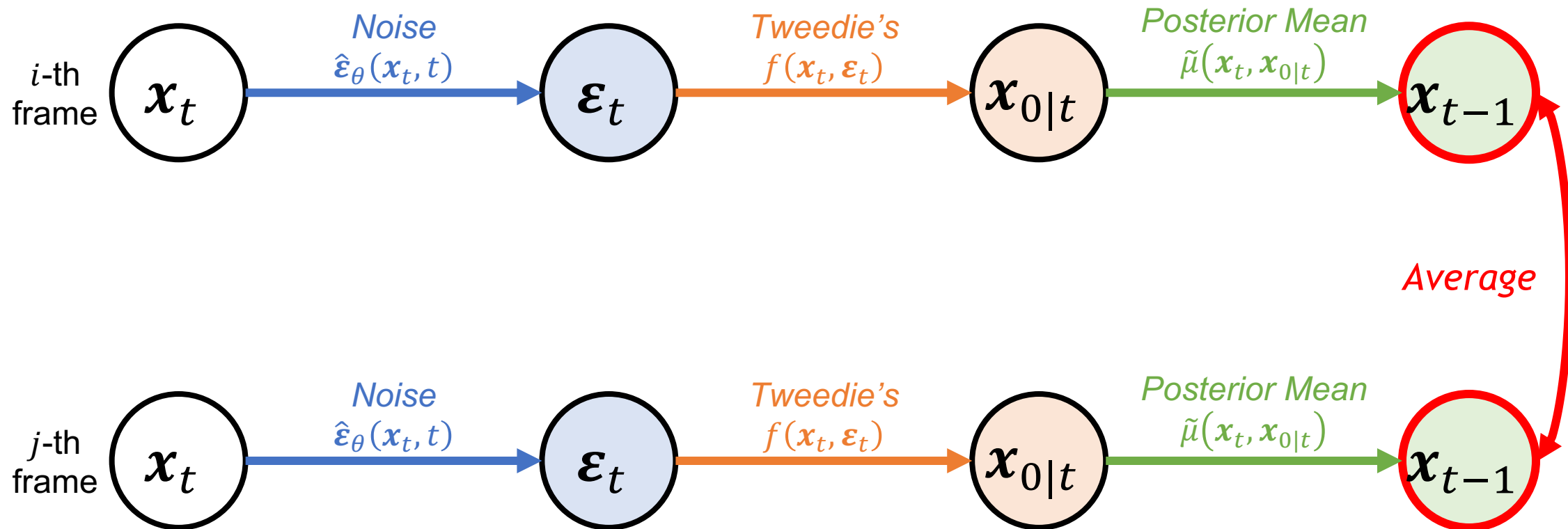
- (+) Generates **seamless** panoramas.
- (-) Images in different windows are **incoherent**.

Coherent Panoramas

The image generated via **LPIPS guidance** is perceptually more **coherent** across frames.

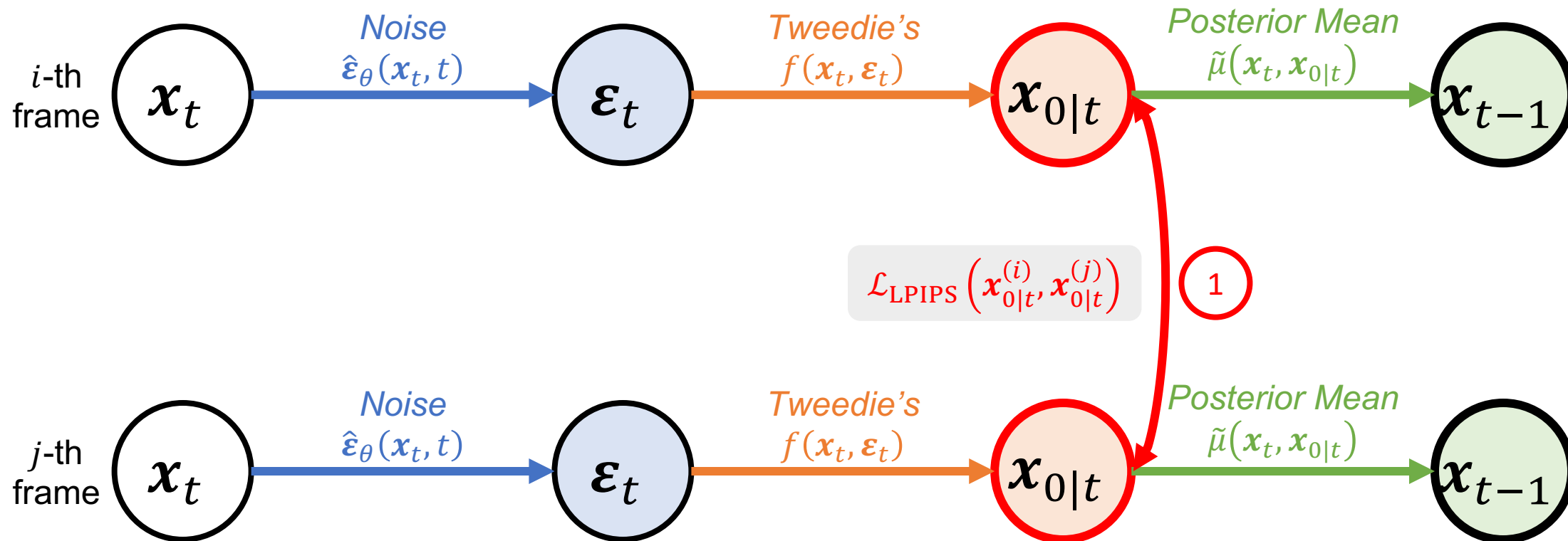


Revisit: MultiDiffusion



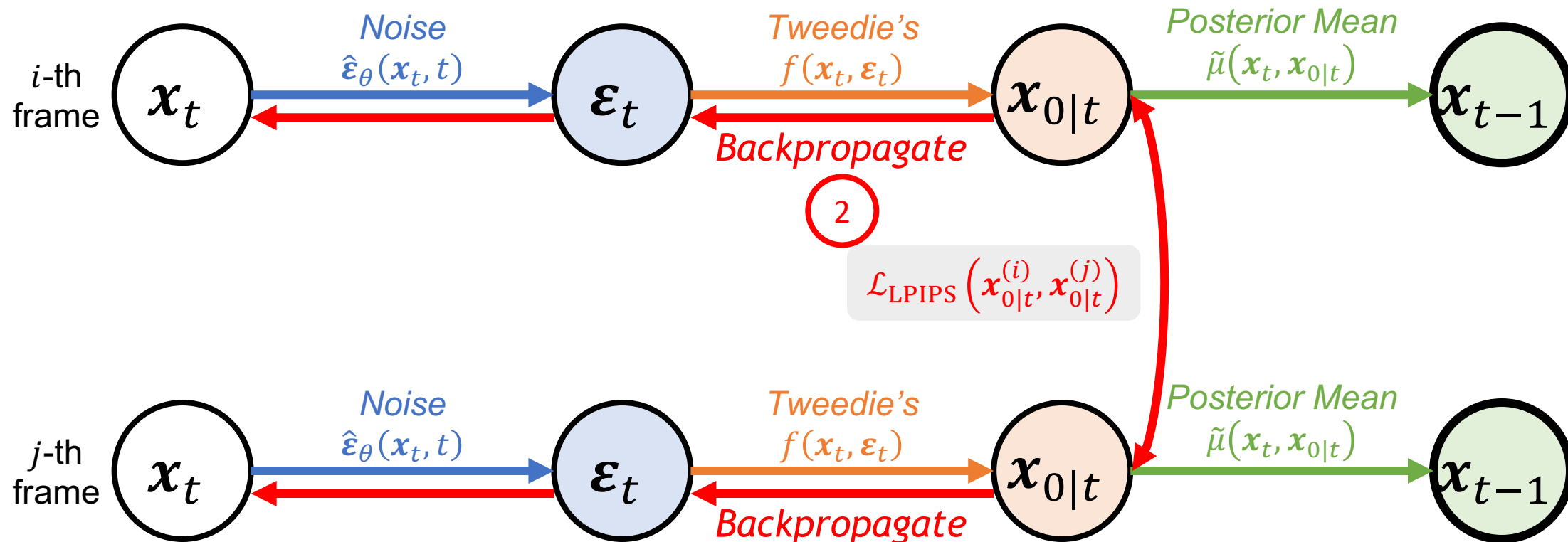
Revisit: MultiDiffusion

Define the guidance function as $\mathcal{L}_{\text{LPIPS}} \left(\mathbf{x}_{0|t}^{(i)}, \mathbf{x}_{0|t}^{(j)} \right)$.



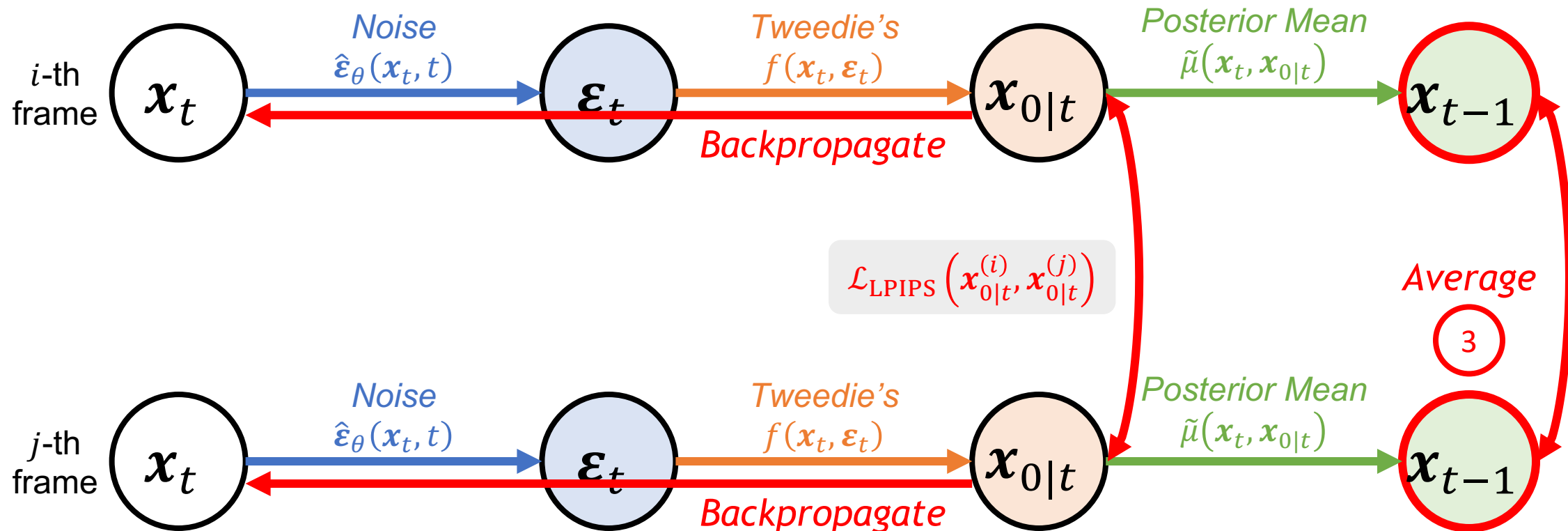
Revisit: MultiDiffusion

Backpropagate from $\mathcal{L}_{\text{LPIPS}} \left(\mathbf{x}_{0|t}^{(i)}, \mathbf{x}_{0|t}^{(j)} \right)$ to $\mathbf{x}_t^{(i)}$ and $\mathbf{x}_t^{(j)}$.



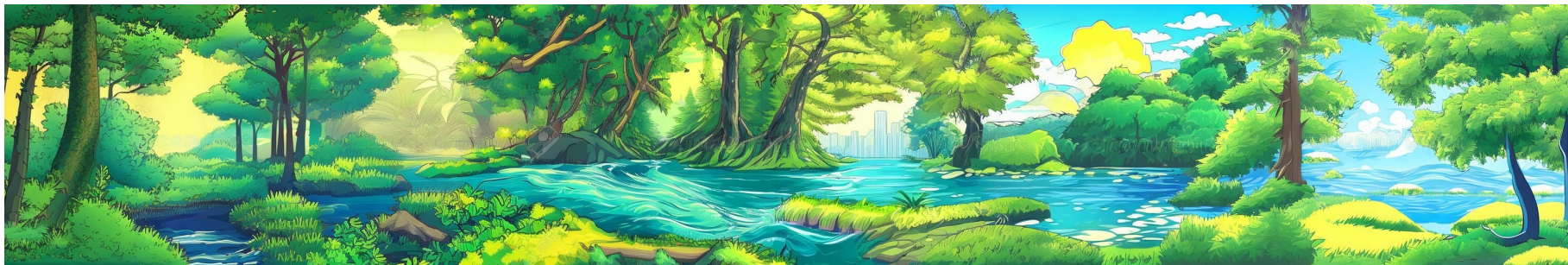
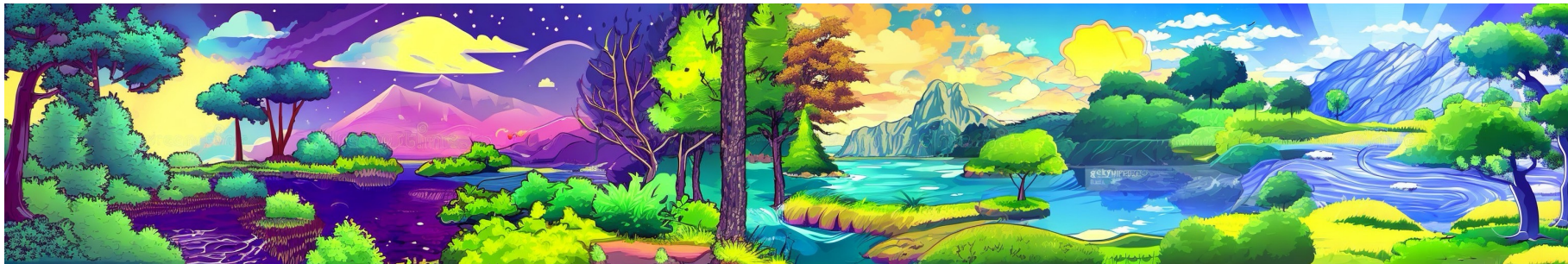
Revisit: MultiDiffusion

Average $\mathbf{x}_{t-1}$.



Coherent Panoramas

The image generated via **LPIPS guidance** is perceptually more **coherent** across frames.



Guided Generation – Summary

- 1. ControlNet**
- 2. Classifier-Free Guidance**
- 3. DPS**
- 4. SDEdit**

Q. 思考以上四種 Guided Generation 招式的使用時機

Guided Generation – Summary

1. Use **ControlNet** if

- it is possible to collect enough condition-output pairs for **finetuning**, and
- when the condition and output share the **same data modality**, and
- when the condition and output are **spatially aligned**.

Guided Generation – Summary

2. Use **Classifier-Free Guidance** if

- it is possible to collect enough condition-output pairs for **finetuning**.
- The condition and output do not need to be spatially aligned or of the same data modality.

Guided Generation – Summary

3. Use **DPS** if

- it is **NOT** possible to collect condition-output pairs for finetuning, or the condition is defined as an **objective function**.
- The objective function must be **differentiable**.

Guided Generation – Summary

4. Use **SDEdit** if

- the relationship between the condition and output is **NOT** specified either in a data-driven way or through an objective function, but
- the goal is to find a realistic data point that closely resembles an unrealistic data point.

Guided Generation – Summary

Technique	ControlNet	Classifier-Free Guidance (CFG)	DPS	SDEdit
Training Data	需要足夠多的 paired training data 來 finetune	需要足夠多的 paired training data 來 finetune	不需要	不需要
Modality	通常 condition 與生成結果是同一種 modality (皆是影像或視訊)	可以處理不同 modalities (text-to-image, audio-to-image)	依據 loss function 而非常彈性	通常 condition 與生成結果是同一種 modality (皆是影像或視訊)
Spatial Alignment	Condition 與生成結果在影像空間中是對齊的	Condition 與生成結果不需要對齊	-	-
Condition 定義	Data-driven (由 training pairs 定義)	Data-driven (由 training pairs 定義)	由 loss function/constraint 定義	沒有明確定義
使用需求	Paired training data with spatial correspondence	Paired training data (any modality/alignment)	Differentiable objective function	Finding realistic version of unrealistic input
使用範例	• Sketch-to-image • Depth-to-image • Pose-to-image • Edge-to-image	• Text-to-image • Novel view synthesis • Class-to-image • Audio-to-video	• Style transfer • Obstacle avoidance • Physics-based constraints • Any differentiable constraint	• Sketch refinement • Image projection to realistic manifold • Coarse-to-fine generation

Guided Generation 使用時機 Decision Flow

1. 你有 **paired training data** 嗎？

- 有：考慮 ControlNet 或 CFG (去 2.)
- 沒有：考慮 DPS 或 SDEdit (去 3.)

2. **Training-based** approaches

- Condition 與生成結果有同樣的 modality 並且在空間上對齊 -> **ControlNet**
- Modalities 不同或空間不對齊 -> **CFG**

3. **Training-free** approaches

- 可以定義/寫出可微分的 loss function -> **DPS**
- Unrealistic to realistic -> **SDEdit**